

## Tutorial 3 – Variables, Constants and Data types

1. Which of the following identifiers are valid? Which are C++ keywords? Give reason(s) if it is invalid.  
  
(miles, Test, a++, --a, 4#R, \$4, #44, apps, double, int, x, y, radius)
2. Find errors if any, from the following declarations and write the correct statements.  
a) Double sum;   b) char marks;   c) int r = 0, s = 0;   d) float payment, NetAmount;
3. In a C++ program declare an integer variable, but while giving input by mistake enter a float value in the variable and display the variable value. Give your comment on the output.
4. In a C++ program declare an unsigned integer variable, but while giving input by mistake enter a negative value in the variable and display the variable value. Give your comment on the output.
5. Write a C++ program to give the input of an integer value greater than 1000. Then multiply the number by itself and display the result.
6. Write a program in C++ that converts kilometers per hour to miles per hour. Prompt the user to enter kilometers/hour and display in miles/ hour. Note that one kilometer is 0.621371 miles.
7. Write a C++ program to find the third angle of a triangle. Create variables to save 1<sup>st</sup> and 2<sup>nd</sup> angle.

Sample Output:

Find the third angle of a triangle:

-----

Input the 1st angle of the triangle: 30

Input the 2nd angle of the triangle: 60

The 3rd of the triangle is: 90

8. (Health application: BMI) Body Mass Index (BMI) is a measure of health on weight. Write a C++ program that prompts the user to enter a weight in pounds and height in inches and displays the BMI. Note that one pound is 0.4536 kilograms and one inch is 0.0254 meters.

**BMI = kg/m<sup>2</sup>**

(use header file `#include<math.h>` and `pow(a, b)` to define  $a^b$ )

sample output:

Enter weight in pounds: 95.5

Enter height in inches: 50

BMI is 26.8577